

- Engineered a fully wireless Raspberry Pi 4B surveillance rover with onboard camera streaming and Bluetooth-based remote control.
- Integrated hardware, software, and mechanical subsystems independently from concept to deployment.
- Achieved approximately 3–4 hours of battery-powered operation and reliable control at a 6-meter line-of-sight range.
- Configured real-time local video streaming for remote monitoring applications.

Manual 3-Axis Agricultural Gantry System

- Led a six-member team in developing a three-axis agricultural gantry prototype for precision irrigation and mechanical weed removal.
 - Designed the motion architecture, wiring, and control methodology while coordinating implementation efforts.
 - Implemented adjustable vertical positioning and targeted water delivery for individual plant treatment.
 - Successfully demonstrated the prototype at a school-level engineering exhibition.
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LEADERSHIP & ACTIVITIES

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